**The due date for submitting this assignment has passed.**
**Due on 2026-03-25, 23:59 IST.**

**You may submit any number of times before the due date. The final submission will be considered for grading.**

You have last submitted on: 2026-03-25, 07:24 IST

**Note: This assignment will be evaluated after the deadline passes. You will get your score 48 hrs after the deadline. Until then the score will be shown as Zero.**

**Multiple Select Questions (MSQ):**

1) A function $T : V \rightarrow W$ between two vector spaces $V$ and $W$ is said to be a linear transformation if the following conditions hold: ***1 point***

**Condition 1:** $T(v_1 + v_2) = T(v_1) + T(v_2)$ for all $v_1, v_2 \in V$.
**Condition 2:** $T(cv) = cT(v)$ for all $v \in V$ and $c \in \mathrm{R}$.

Consider the following function:

$$T : \mathrm{R}^2 \rightarrow \mathrm{R}$$
$$T(x, y) = \begin{cases} 3x & \text{if } y = 0 \\ 4y & \text{if } y = 0 \end{cases}$$

Which of the following statements is true?

- [ ] Condition 1 holds.
- [x] Condition 1 does not hold.
- [x] Condition 2 holds.
- [ ] Condition 2 does not hold.

Yes, the answer is correct.

Score: 1

Accepted Answers:

Condition 1 does not hold.
Condition 2 holds.

2) Suppose the matrix representation of a linear transformation $T : \mathrm{R}^3 \rightarrow \mathrm{R}^3$ with respect to ordered bases $\beta = \{(1, 0, 1), (0, 1, 0), (0, 0, 1)\}$ for the domain and $\gamma = \{(1, 0, 0), (0, 1, 0), (1, 0, 1)\}$ for the range, is $I_{3\times3}$, i.e., the identity matrix of order 3. Let $A$ denote the matrix representation of the linear transformation $T$ with respect to the standard ordered basis of $\mathrm{R}^3$ for both domain and range. Which of the following are true? ***1 point***

- [ ] $A = I_{3\times3}$ i.e., identity matrix of order 3.
- [ ] $A$ is a singular matrix
- [ ]

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & -1 \end{bmatrix}$$

- [x] $det(A) = 1.$
- [ ] $det(A) = -1.$
- [x] $A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$

Yes, the answer is correct.

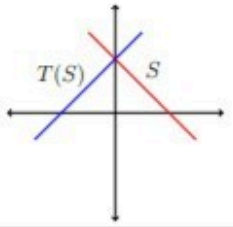
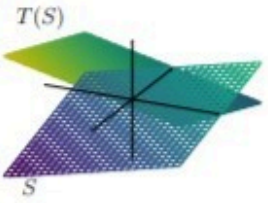
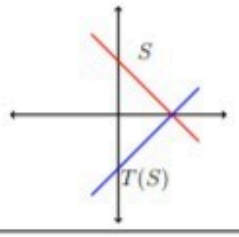
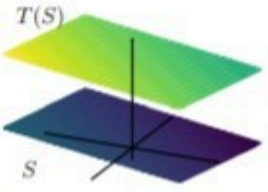
Score: 1

Accepted Answers:

$det(A) = 1.$

$A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$

3) Match the linear transformations and sets of vectors in column A with the images of those sets under the linear transformation in column B and the geometric representations of both sets in column C. ***1 point***

| | Matrix form of linear transformation (Column A) | | Image of the given set (Column B) | | Geometric representations (Column C) |
|---|---|---|---|---|---|
| i) | $T : \mathbb{R}^3 \to \mathbb{R}^3$ $T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 2 & 3 \end{bmatrix}$ Set: $S = \{(x, y, z) \mid x + y + z = 1\}$ | a) | $T(S) = \{(x, y) \mid x - y = 1\}$ | 1) |   |
| ii) | $T : \mathbb{R}^3 \to \mathbb{R}^3$ $T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ Set: $S = \{(x, y, z) \mid x + y + z = 1\}$ | b) | $T(S) = \{(x, y, z) \mid x + y + z = 3\}$ | 2) |   |
| iii) | $T : \mathbb{R}^2 \to \mathbb{R}^2$ $T = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ Set: $S = \{(x, y) \mid x + y = 1\}$ | c) | $T(S) = \{(x, y) \mid x - y = -1\}$ | 3) |   |
| iv) | $T : \mathbb{R}^2 \to \mathbb{R}^2$ $T = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$ Set: $S = \{(x, y) \mid x + y = 1\}$ | d) | $T(S) = \{(x, y, z) \mid x + y - z = 1\}$ | 4) |   |

Choose the correct option from the following.

- [ ] i → d → 2, ii → b → 4.

i → b → 4, ii → d → 2.

- [x] iii → a → 1, iv → c → 3
- [ ] iii → a → 3, iv → c → 1

No, the answer is incorrect.

Score: 0

Accepted Answers:

i → b → 4, ii → d → 2.
iii → a → 3, iv → c → 1

4) Consider a linear transformation $S : M_2(\mathrm{R}) \rightarrow M_2(\mathrm{R})$ such that $S(A) = A^T$. Let $B$ ***1 point*** be the matrix representation of $S$ with respect to the ordered bases:

$$\{\begin{bmatrix}1 & 0\\0 & 0\end{bmatrix}, \begin{bmatrix}0 & 1\\0 & 0\end{bmatrix}, \begin{bmatrix}0 & 0\\1 & 0\end{bmatrix}, \begin{bmatrix}0 & 0\\0 & 1\end{bmatrix}\}$$

of $M_2(\mathrm{R})$. Choose the set of correct options:

- [ ] The order of the matrix $B$ is $2 \times 2$.
- [x] The order of the matrix $B$ is $4 \times 4$.
- [x] The dimension of the row space of the the matrix $B$ is 4.
- [ ] The dimension of the column space of the matrix $B$ is 3.
- [ ] The nullity of the the matrix $B$ is 1.
- [x] The rank of the the matrix $B$ is 4.
- [x] $S$ is surjective.

Yes, the answer is correct.

Score: 1

Accepted Answers:

The order of the matrix $B$ is $4 \times 4$.
The dimension of the row space of the the matrix $B$ is 4.
The rank of the the matrix $B$ is 4.

$S$ is surjective.

5) Let $T : \mathrm{R}^n \rightarrow \mathrm{R}^m$ be a linear transformation and tet $\{v_1, v_2, \ldots, v_n\}$ be a basis of $\mathrm{R}^n$. Suppose $\{T(v_1), T(v_2) \ldots, T(v_n)\}$ is a spanning set for $\mathrm{R}^m$. Choose all the correct statements. ***1 point***

- [x] $T$ is a surjective linear transformation.
- [ ] $T$ is a linear isomorphism.
- [x] $m \leq n$.
- [x] For any basis $\{w_1, w_2, \ldots, w_n\}$ of $\mathrm{R}^n$, the set $\{T(w_1), T(w_2), \ldots, T(w_n)\}$ is a spanning set for $\mathrm{R}^m$.
- [ ] The matrix of the linear transformation $T$ has rank equal to $n$.

Yes, the answer is correct.

Score: 1

Accepted Answers:

$T$ is a surjective linear transformation.
$m \leq n$.
For any basis $\{w_1, w_2, \ldots, w_n\}$ of $\mathrm{R}^n$, the set $\{T(w_1), T(w_2), \ldots, T(w_n)\}$ is a spanning set for $\mathrm{R}^m$.

**Comprehension Type Question:**

Suppose a bread-making machine B makes 6 breads from 2 eggs, 3 (in hundreds) grams of wheat, and 1 (in hundred) grams of sugar. B also makes 8 breads from 3 eggs, 4 (in hundreds) grams of wheat, and 2 (in hundreds) grams of sugar, and 10 breads from 5 eggs, 5 (in hundreds) grams of wheat, and 3 (in hundreds) grams of sugar. Suppose the production of breads is a linear function of the amount of eggs, wheat (in hundreds), and sugar (in hundreds) used as raw ingredients. Based on the above data answer the following questions. Suppose $x$ eggs, $y$ (in hundreds) grams of wheat, and $z$ (in hundreds) grams of sugar are used as the raw materials to produce $ax + by + cz$ number of breads. We can express this as follows:

$$T : \mathrm{R}^3 \rightarrow \mathrm{R}$$

$$T(x, y, z) = ax + by + cz$$

where the co-ordinates in $\mathrm{R}^3$ denote the number of eggs, amount (in grams) of wheat (in hundreds), and amount (in grams) of sugar (in hundreds). Observe that $T$ is a linear transformation.

**Use the above information for Question 6, 7 and 8.**

6) Choose the correct set of options from the the following. (MSQ) ***1 point***

- [ ] $Nullity(T) = 1$
- [x] $Rank(T) = 1$
- [x] $Nullity(T) = 2$
- [ ] $Rank(T) = 2$
- [ ] $Nullity(T) = 3$
- [ ] $Rank(T) = 3$
- [ ] $T$ is neither one to one nor onto.
- [ ] $T$ is one to one but not onto.
- [x] $T$ is onto but not one to one.
- [ ] $T$ is an isomorphism.

Yes, the answer is correct.

Score: 1

Accepted Answers:

$Rank(T) = 1$
$Nullity(T) = 2$
$T$ is onto but not one to one.

7) Choose the set of correct statements. (MSQ) ***1 point***

- [ ] If 4 eggs and 2 (in hundreds) grams of sugar is used, and no wheat is used, then 9 breads are produced.
- [x] If 4 eggs and 2 (in hundreds) grams of sugar is used, and no wheat is used, then no bread is produced.
- [x] If only 3 (in hundreds) grams of wheat is used, then 6 breads are produced.

- [ ] If 3 (in hundreds) grams of wheat and 1 (in hundred) grams of sugar is used, and no egg is used, then no bread is produced.
- [x] If 3 (in hundreds) grams of wheat and 1 (in hundred) grams of sugar is used, and no egg is used, then 6 breads are produced.

Yes, the answer is correct.

Score: 1

Accepted Answers:

If 4 eggs and 2 (in hundreds) grams of sugar is used, and no wheat is used, then no bread is produced.
If only 3 (in hundreds) grams of wheat is used, then 6 breads are produced.
If 3 (in hundreds) grams of wheat and 1 (in hundred) grams of sugar is used, and no egg is used, then 6 breads are produced.

8) How many breads are produced by the machine from 6 eggs, 10 (in hundreds) grams of wheat, and 5 (in hundreds) grams of sugar? (NAT)

20

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 20

***1 point***

9) Let $T$ be a linear transformation from $\mathrm{R}^{10}$ to $\mathrm{R}^{12}$. Suppose a basis for the null space of $T$ has $3$ vectors. How many linearly independent vectors are needed to form a basis for the range of $T$?

7

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 7

***1 point***

10) Let $\mathrm{M}$ be the set of all skew-symmetric matrices of order 7. Then $\mathrm{M}$ forms a vector space under matrix addition and scalar multiplication. Let $T$ be a linear transformation from $\mathrm{M}$ to $\mathrm{R}$ defined by $T(A) = 6\ tr(A)$, where $tr(A)$ =trace of $A$. What is the nullity of $T$?

21

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 21.0

***1 point***

11) Let $T : \mathrm{R}^3 \longrightarrow \mathrm{R}^2$ be defined by $T(x, y, z) = (10x + 9y, 2z)$. Choose the correct options about $T$. ***1 point***

- [ ] The matrix of $T$ is $\begin{bmatrix} 10 & 0 \\ 9 & 0 \\ 0 & 2 \end{bmatrix}$.
- [x] The matrix of $T$ is $\begin{bmatrix} 10 & 9 & 0 \\ 0 & 0 & 2 \end{bmatrix}$.
- [x] A basis for the null space of $T$ is $\{(1, -\frac{10}{9}, 0)\}$.
- [ ] A basis for the null space of $T$ is $\{(1, -\frac{10}{9}, 0), (0, 0, 2)\}$.
- [ ] A basis for the range of $T$ is $\{(10, 10)\}$.
- [x] A basis for the range of $T$ is $\{(10, 9), (0, 2)\}$.

Yes, the answer is correct.

Score: 1

Accepted Answers:

The matrix of $T$ is $\begin{bmatrix} 10 & 9 & 0 \\ 0 & 0 & 2 \end{bmatrix}$.

A basis for the null space of $T$ is $\{(1, -\frac{10}{9}, 0)\}$.

A basis for the range of $T$ is $\{(10, 9), (0, 2)\}$.